Group(8)_CW1_Foyaz_Ahmed_w2140035

Individual Contribution Essay

For our group coursework, we decided to develop a LEGO themed pizza delivery simulator using Unity. The main idea is players receive a customer order, collect a pizza, travel around a small town, deliver to the correct house and receive a score before moving on to the next order. I worked on different areas so the project has a clear design and technical components can be tested. This essay includes what I have personally contributed and supported the group as well as the parts that need improvement.

One of my first tasks was to help organise the group will work, we had a meeting on 10 October 2026 , we had a talk to set up the project and GitHub and delegated different responsibilities to each other fairly. We also created a whatsapp group so that we could communicate on a daily basis and keep effective communication, and we had a shared OneDrive folder for storing all the work we have done which was good and helpful for us . I also had another responsibility in keeping up with the meetings and schedules as well as interaction design and diagrams. The other members took responsibility for Git, presentations and administration and everyone also came to a agreement to contribute the assets and codings and animations. I also kept record of meetings so we could refer back to decisions and so that all our responsibilities were clear so that our work is kept organized instead of going off of conversations that we dont remember.

We held our second meeting on 13 October 2026 where we focused on the design of the environment and potential gameplay features. Most of my contribution went to planning the positions of roads, traffic lights, greenery and obstacles. To make the deliveries abit more challenging we was going to add in road closures or obstacles and had also discussed a system where quicker deliveries could earn better tips. We also had a discussion about adding customer feedback and adding a phone style interface which shows info such as maps, speed and communication with customers. These ideas helped explore more possibilities for our game even though some were just potential proposals rather than finalised features. I still took note of the discussions so we had a record of which direction we wanted to take.

I also made an early concept layout of the LEGO delivery area which shows a pizza shop, houses A,B and C, a park or props area and crossroads arrangement with a central junction. However it isn't drawn to scale but it did help visualize routes a player can possibly take between the pizza shop and also the customers. This also gave us a starting point to talk as a group on where scenery and obstacles could potentially be placed in the design. Therefore, preparing the drawing before building the full level was quite useful as it gave the group a shared reference and something to work from and build up on for the shape and purpose of the environment.

Another big part of my contribution to the group was designing the gameplay flow, I created flow charts that explained how the pizza delivery process worked or should work beginning with, receving an order, collecting the pizza, navigating the customers, delicvering it, calculating the score and then of course receiving the next order. I have also included a arrow link to show the final stage all the way back to the first stage so that it shows the repated gameplay loop as it makes it much easier to show and explain the overall experience and identify how all the individual systems must work together for the gameplay to be successful. In addition to the gameplay flowchart I have also prepared a basic use-case diagram which shows every single players intended interactions such as the movememt as well as the camera control, collecting and delivering pizzas, viewing points, checking the scored and the timer, receiving new orders and also pausing or possibly restarting the game.

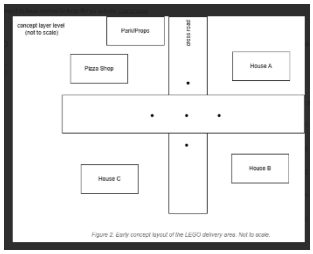


Figure 2. Early concept layout of the LEGO delivery area. Not to scale.

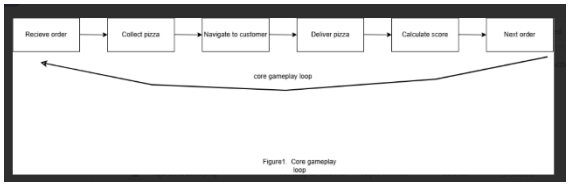


Figure1. Core gameplay loop

I have also made a diagram using draw.io showing the planned relationships between the unity scripts . This included PlayMovement, CameraController, GameManager , PizzaPickup , DeliveryPoint, DeliveryTimer, ScoreManager and UIManager. The diagram was made also so it can explain which scripts would communicate with one another during the delivery process. For example completeing a delivery should affect scoring and timing and the results should be displayed on the user interface. It is helpful and good for understanding how the work could be divided between us all. The diagram represents our structure .

In meeting three on 20th october 2026 the group allocated the basic controls and physics tasks from tutorials 02 and 03 and I chose to do the physics controller while other members assigned the character controller , mouse input controller and physics wall. For my individual task I worked with a LEGO brick and cube in the unity and added a rigbody component and developed a physics controller.cs script . I used the tutorial and lecture on blaclboard to help me as well as youtube tutorials on the parts I struggled with. The controller used keyboard input WASD and space as part of the movement setup.

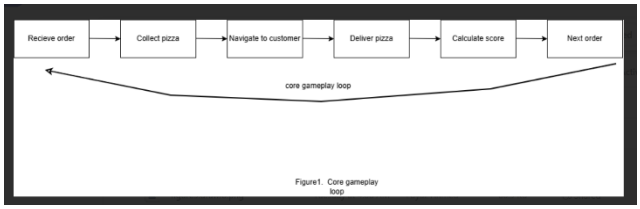


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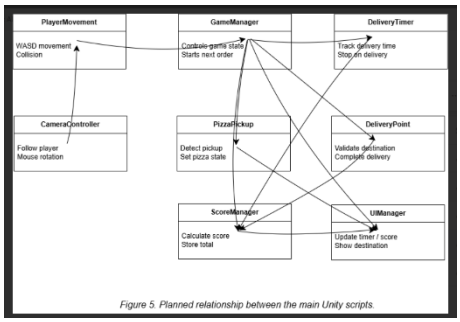


Figure 5. Planned relationship between the main Unity scripts.

i also contributed to the Unity scene setup and presentation. During the development I worked on the envireonment containing basic roads and ground and looked at the gamerplay view with a score , and countdown timer and delivery destination label.this helped me understand how interface info needs to be presented while the player is moving around and walking and the level. I created a separate Main menu scene and added 2 buttons labelled Play Game and Quit Game. I then worked on a MainMenu c# script for the button actions .

Another important practical task was importing and configuring a LEGO minifigure with animation. I imported the character and animation files into Unity from the blackboard , added the minifigure to a scene and set up an Animator Controller named WAC. Inside the Animator I created standing and walking states, connected them through transitions and added a PlayWalk trigger. I also created a WalkButton.cs script containing a method that sends this trigger to the character's Animator. After adding UI button I changed its text to Play Walk and connected its On Click event to the PlayWalk method. This successfully linked a user interface action to the animation controller which is the central requirement of the minifigure animation activity.

When testing the animation it showed a problem that I have been trying to figure out. When I clicked Play Walk the character moved very quickly so it disappeared from the camera view and then returned back very quickly when it sould have been staying in place and walking normally. I tried reducing the animation speed as I thought it was too quick and checked that Apply Root Motion was unticked. I also examined the standing and walking states and found a state labelled walk (1) had been assigned a Flip motion which I done mistakingly

while following the blackboard video . I corrected the assigned motion and inspected the imported clips and their position and rotation settings. The button is linked to the Animator, I cant yet confirm that the animation plays correctly in the game scene. This troubleshooting taught me that an animation preview can look correct even when its behaviour in the full scene is not.

I worked on positioning the camera behind the LEGO minifigure to create a basic third-person view. By adjusting the Scene view and aligning the Main Camera to it I made the character visible in the Game view. Then the scene had a more suitable perspective for controlling a character around a delivery environment and looked more better and normal. But a complete follow camera system hasnt been fully confirmed especially when the character moves or plays animations. My main focus was to make a clear view that can be developed further as the control system improves.

In my contribution I have taken part in both the design and development stages of the project. Writing meeting records and organising schedules supported communication and the level layout and diagrams helped the group explain the intended game and divide up the technical responsibilities. My Unity work included the Rigidbody physics controller and menu interface and LEGO minifigure and animation controller and the button scripting and third-person camera setup. Also those tasks have helped me improve my understanding of Unity components and C# scripts and the Animator triggers and the user interface events. Before submission, I need to finish investigating the character's unwanted movement and verify the physics controller and menu buttons. In general my contribution has helped establish the design documentation and practical foundations for our LEGO Pizza Delivery Simulator.

I mainly used the tutorials to help me with this coursework especially the Rigbody phyiscs , MainMenu , the Lego Minifigure and the animations etc and used youtube videeos on when I was struggling for the walking animation when the Minifigure would walk really fast and disappear and appear back I was struggling to find the issue for it and the solutions for it so I asked to AI to help show me where the problem is and what the problem is.